MALEKA SHELLTON

JUNIOR DATA SCIENTIST

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Junior Data Scientist completing a Postgraduate Diploma in Data Science at Wits, with hands-on experience in machine learning, NLP, LLMs and big-data analytics. Proficient in Python, SQL and R, with practical experience building retrieval-augmented generation (RAG) applications using LangChain, Chroma and locally hosted LLMs, distributed ML with Apache Spark, and end-to-end model deployment.

EDUCATION

POSTGRADUATE DIPLOMA IN DATA SCIENCE **02/2026 – PRESENT**
University of the Witwatersrand | Johannesburg

BACHELOR OF EDUCATION – Mathematics & Physical Sciences **02/2022 – 12/2025**
University of the Witwatersrand | Johannesburg

TECHNICAL SKILLS

Programming & Data: Python, SQL, R
Machine Learning: Scikit-learn, classification, regression, feature engineering, model evaluation
NLP & Generative AI: LangChain, RAG, Sentence Transformers, Chroma, Ollama, LLMs
Big Data: Apache Spark, Spark MLlib
Tools & Deployment: Git, GitHub, Streamlit
Analytics: Power BI, Excel, statistical analysis

SELECTED DATA SCIENCE PROJECTS

Data Science Study Assistant – RAG & Large Language Models | GitHub
Python | LangChain | Ollama | Chroma | Sentence Transformers | Streamlit
Built a RAG application for querying data science learning materials in natural language, using semantic embeddings, Chroma vector retrieval and a locally hosted Ollama LLM to generate grounded responses.

Customer Churn Prediction – End-to-End ML Application | GitHub
Python | Scikit-learn | Streamlit | Joblib
Built an end-to-end classification pipeline on 10,000 customer records with preprocessing, multi-model comparison and evaluation using accuracy, precision, recall and F1-score; deployed real-time churn probabilities with Streamlit.

Income Classification using Apache Spark MLlib | GitHub
Python | PySpark | Spark MLlib | Distributed Machine Learning
Implemented a distributed workflow covering preprocessing, EDA, feature engineering, model training, evaluation and interpretation, demonstrating scalable machine-learning pipeline development.

CERTIFICATIONS

Microsoft Certified: Azure Fundamentals (AZ-900) | AWS Educate Machine Learning Foundations | Cisco Data Analytics Essentials | BCG X Data Science Job Simulation

LEADERSHIP & EXPERIENCE

Residence Maintenance Officer | University of the Witwatersrand **02/2026 – PRESENT**
Coordinate resident maintenance reporting, unresolved-issue follow-up and operational communication between residents and management.

First Year Experience Mentor | University of the Witwatersrand DLU **02/2023 – 11/2023**
Guided first-year students through academic transition and communicated support information to diverse audiences.

AWARDS & LANGUAGES

Golden Key Southern Africa Award (2023) | Languages: Tshivenda, IsiZulu, English